

## Experiment 5: Cloud Object Storage Manager Using Python and MinIO

### Aim

To implement a Python-based cloud object storage manager using a local MinIO S3-compatible server, supporting bucket creation, upload, download, listing, metadata display, and object deletion.

### Tools Required

- Ubuntu VM
- Python 3
- MinIO Community Server
- MinIO Python SDK
- Python virtual environment
- Browser for the MinIO console

### Detailed Procedure

**Update Ubuntu and install Python tools.**

```
sudo apt update
sudo apt install python3 python3-pip python3-venv golang-go -y
```

**Create a directory for the MinIO source and build/install the Community server. The prebuilt legacy download route was not used because it returned an unavailable-resource error.**

```
mkdir ~/minio
cd ~/minio
go install github.com/minio/minio@latest
sudo mkdir -p /usr/local/bin
sudo cp ~/go/bin/minio /usr/local/bin/minio
minio --version
```

**Create a clean storage directory.**

```
mv ~/minio-data ~/minio-data-old 2>/dev/null || true
mkdir ~/minio-data
```

**Set the MinIO root credentials.**

```
export MINIO_ROOT_USER=minioadmin
export MINIO_ROOT_PASSWORD=minioadmin
```

**Start the MinIO server. Keep this terminal running while using the storage application.**

```
minio server ~/minio-data --console-address ":9001"
```

**Open the MinIO console in a browser at <http://127.0.0.1:9001/browser> and sign in using minioadmin / minioadmin.**

**In a second terminal, create the Python project and virtual environment.**

```
mkdir ~/ObjectStorageSimulator
cd ~/ObjectStorageSimulator
python3 -m venv venv
source venv/bin/activate
```

### Install the official MinIO Python client.

```
pip install minio
```

### Create a sample file.

```
echo "Hello this is sample minio project" > sample.txt
```

**Create a Python manager program that connects to localhost:9000 and uses the bucket studentbucket. Implement create bucket, upload, download, list objects with size/last-modified metadata, and delete object operations.**

**Run the manager and verify each operation. The server terminal must remain active; otherwise the client will report a connection-refused error.**

### Core Manager Code

```
from minio import Minio
from minio.error import S3Error

client = Minio("localhost:9000", access_key="minioadmin", secret_key="minioadmin", secure=False)
BUCKET = "studentbucket"

def create_bucket():
    if not client.bucket_exists(BUCKET):
        client.make_bucket(BUCKET); print("Bucket Created")
    else:
        print("Bucket Already Exists")

def upload_object():
    client.fput_object(BUCKET, "notes.txt", "sample.txt"); print("Upload Successful")

def download_object():
    client.fget_object(BUCKET, "notes.txt", "downloaded.txt"); print("Download Successful")

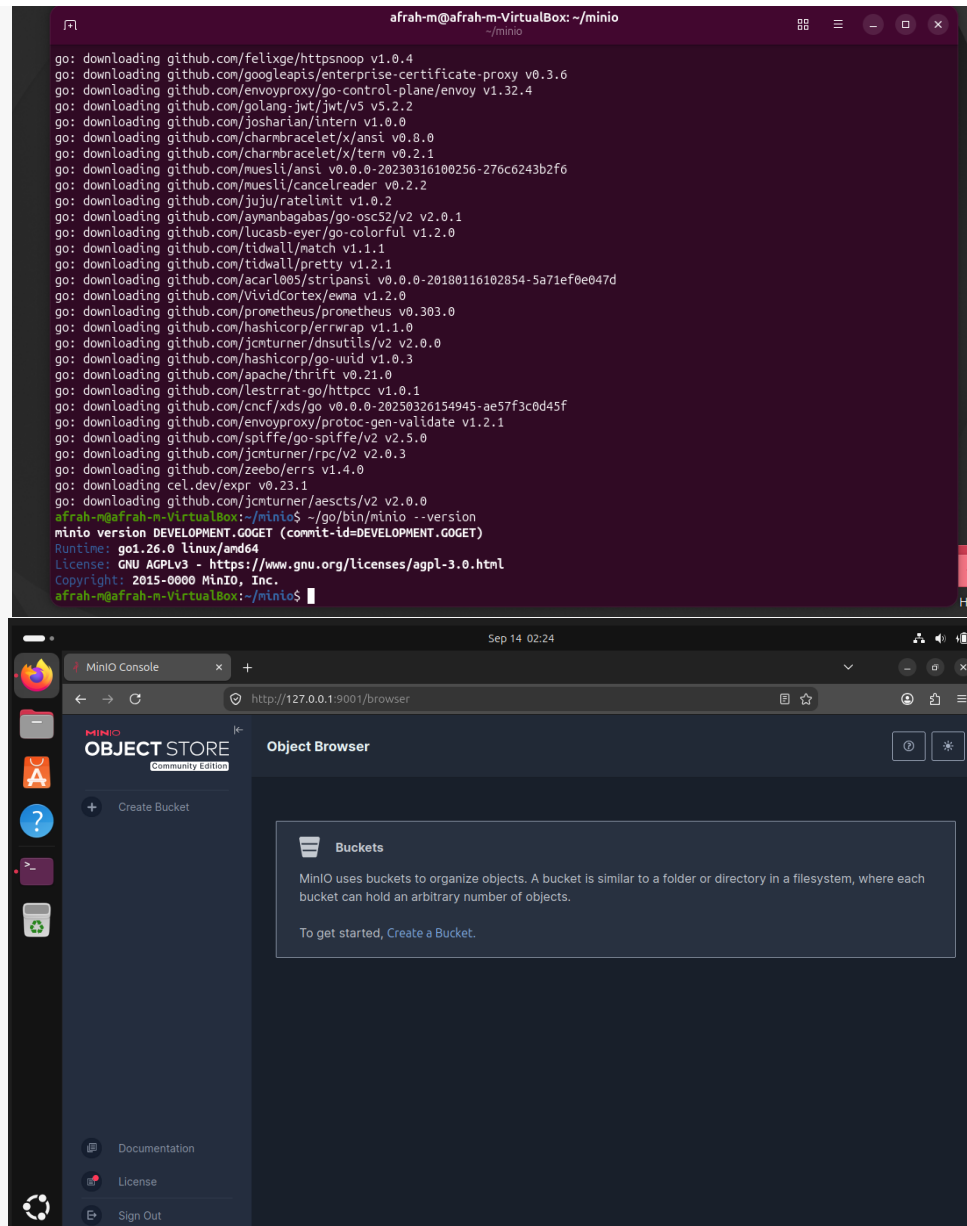
def list_objects():
    print("\nBucket Contents:")
    found = False
    for obj in client.list_objects(BUCKET):
        found = True
        print("Object:", obj.object_name)
        print("Size:", obj.size, "bytes")
        print("Last Modified:", obj.last_modified)
        print("-" * 40)
    if not found: print("Bucket is empty.")

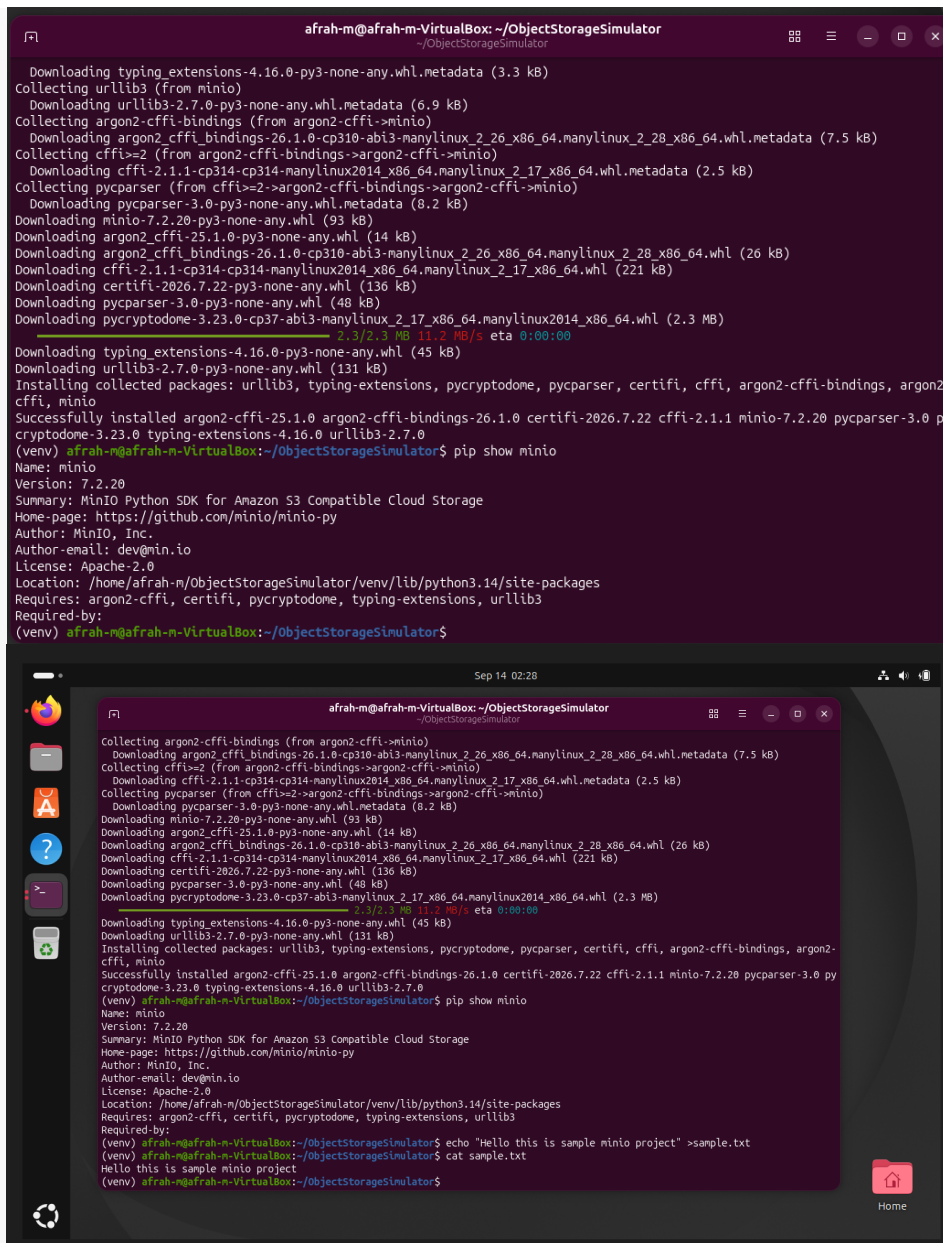
def delete_object():
    client.remove_object(BUCKET, "notes.txt"); print("Object Deleted Successfully")

while True:
    print("\n==== Object Storage Manager =====")
    print("1. Create Bucket\n2. Upload Object\n3. Download Object\n4. List Objects\n5. Delete Object\n6. Exit")
    choice = input("Enter your choice: ")
    try:
        if choice == "1": create_bucket()
        elif choice == "2": upload_object()
```

```
elif choice == "3": download_object()
elif choice == "4": list_objects()
elif choice == "5": delete_object()
elif choice == "6": print("Exiting..."); break
else: print("Invalid choice.")
except S3Error as e: print("MinIO Error:", e)
```

## Output





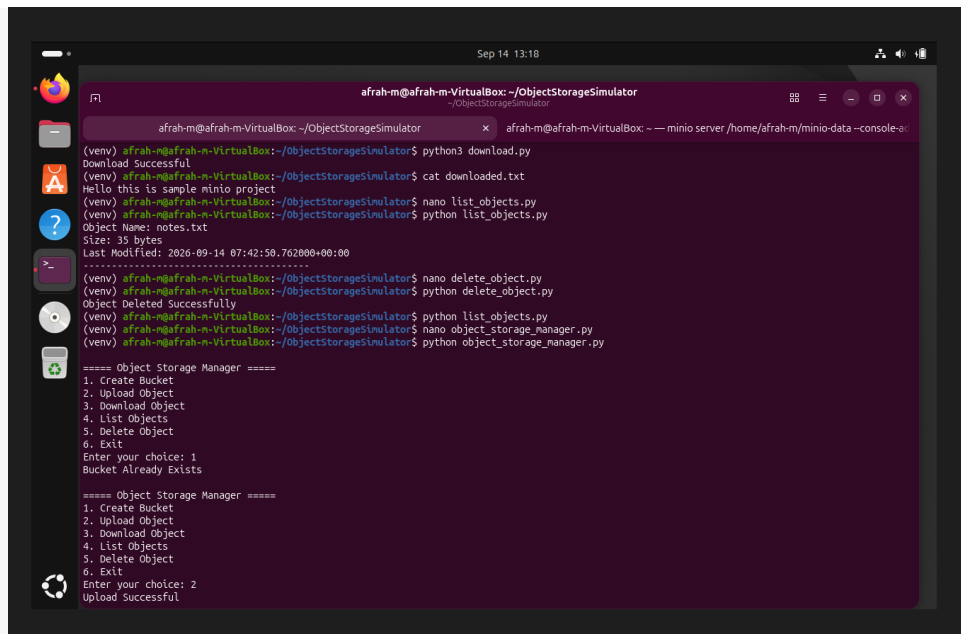
```
afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator
~/ObjectStorageSimulator

Downloading typing_extensions-4.16.0-py3-none-any.whl.metadata (3.3 kB)
Collecting urllib3 (from minio)
  Downloading urllib3-2.7.0-py3-none-any.whl.metadata (6.9 kB)
Collecting argon2-cffi-bindings (from argon2-cffi->minio)
  Downloading argon2_cffi_bindings-26.1.0-cp310-abi3-manylinux_2_26_x86_64.manylinux_2_28_x86_64.whl.metadata (7.5 kB)
Collecting cffi>=2 (from argon2-cffi-bindings->argon2-cffi->minio)
  Downloading cffi-2.1.1-cp314-cp314-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (2.5 kB)
Collecting pycparser (from cffi>=2->argon2-cffi-bindings->argon2-cffi->minio)
  Downloading pycparser-3.0-py3-none-any.whl.metadata (8.2 kB)
Downloading minio-7.2.20-py3-none-any.whl (93 kB)
Downloading argon2_cffi_bindings-25.1.0-py3-none-any.whl (14 kB)
Downloading argon2_cffi_bindings-26.1.0-cp310-abi3-manylinux_2_26_x86_64.manylinux_2_28_x86_64.whl (26 kB)
Downloading cffi-2.1.1-cp314-cp314-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (221 kB)
Downloading certifi-2026.7.22-py3-none-any.whl (136 kB)
Downloading pycparser-3.0-py3-none-any.whl (48 kB)
Downloading pycryptodome-3.23.0-cp37-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.3 MB)
2.3/2.3 MB 11.2 MB/s eta 0:00:00
Downloading typing_extensions-4.16.0-py3-none-any.whl (45 kB)
Downloading urllib3-2.7.0-py3-none-any.whl (131 kB)
Installing collected packages: urllib3, typing-extensions, pycryptodome, pycparser, certifi, cffi, argon2-cffi-bindings, argon2-cffi, minio
Successfully installed argon2-cffi-25.1.0 argon2-cffi-bindings-26.1.0 certifi-2026.7.22 cffi-2.1.1 minio-7.2.20 pycparser-3.0 py
cryptodome-3.23.0 typing-extensions-4.16.0 urllib3-2.7.0
(venv) afrah-m@afrah-m-VirtualBox:~/ObjectStorageSimulator$ pip show minio
Name: minio
Version: 7.2.20
Summary: MinIO Python SDK for Amazon S3 Compatible Cloud Storage
Home-page: https://github.com/minio/minio-py
Author: MinIO, Inc.
Author-email: dev@min.io
License: Apache-2.0
Location: /home/afrah-m/ObjectStorageSimulator/venv/lib/python3.14/site-packages
Requires: argon2-cffi, certifi, pycryptodome, typing-extensions, urllib3
Required-by:
(venv) afrah-m@afrah-m-VirtualBox:~/ObjectStorageSimulator$

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afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator
~/ObjectStorageSimulator

Collecting argon2-cffi-bindings (from argon2-cffi->minio)
  Downloading argon2_cffi_bindings-26.1.0-cp310-abi3-manylinux_2_26_x86_64.manylinux_2_28_x86_64.whl.metadata (7.5 kB)
Collecting cffi>=2 (from argon2-cffi-bindings->argon2-cffi->minio)
  Downloading cffi-2.1.1-cp314-cp314-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (2.5 kB)
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  Downloading pycparser-3.0-py3-none-any.whl.metadata (8.2 kB)
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Downloading argon2_cffi_bindings-25.1.0-py3-none-any.whl (14 kB)
Downloading argon2_cffi_bindings-26.1.0-cp310-abi3-manylinux_2_26_x86_64.manylinux_2_28_x86_64.whl (26 kB)
Downloading cffi-2.1.1-cp314-cp314-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (221 kB)
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Downloading pycryptodome-3.23.0-cp37-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.3 MB)
2.3/2.3 MB 11.2 MB/s eta 0:00:00
Downloading typing_extensions-4.16.0-py3-none-any.whl (45 kB)
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Installing collected packages: urllib3, typing-extensions, pycryptodome, pycparser, certifi, cffi, argon2-cffi-bindings, argon2-cffi, minio
Successfully installed argon2-cffi-25.1.0 argon2-cffi-bindings-26.1.0 certifi-2026.7.22 cffi-2.1.1 minio-7.2.20 pycparser-3.0 py
cryptodome-3.23.0 typing-extensions-4.16.0 urllib3-2.7.0
(venv) afrah-m@afrah-m-VirtualBox:~/ObjectStorageSimulator$ pip show minio
Name: minio
Version: 7.2.20
Summary: MinIO Python SDK for Amazon S3 Compatible Cloud Storage
Home-page: https://github.com/minio/minio-py
Author: MinIO, Inc.
Author-email: dev@min.io
License: Apache-2.0
Location: /home/afrah-m/ObjectStorageSimulator/venv/lib/python3.14/site-packages
Requires: argon2-cffi, certifi, pycryptodome, typing-extensions, urllib3
Required-by:
(venv) afrah-m@afrah-m-VirtualBox:~/ObjectStorageSimulator$ echo "Hello this is sample minio project" >sample.txt
(venv) afrah-m@afrah-m-VirtualBox:~/ObjectStorageSimulator$ cat sample.txt
Hello this is sample minio project
(venv) afrah-m@afrah-m-VirtualBox:~/ObjectStorageSimulator$
```

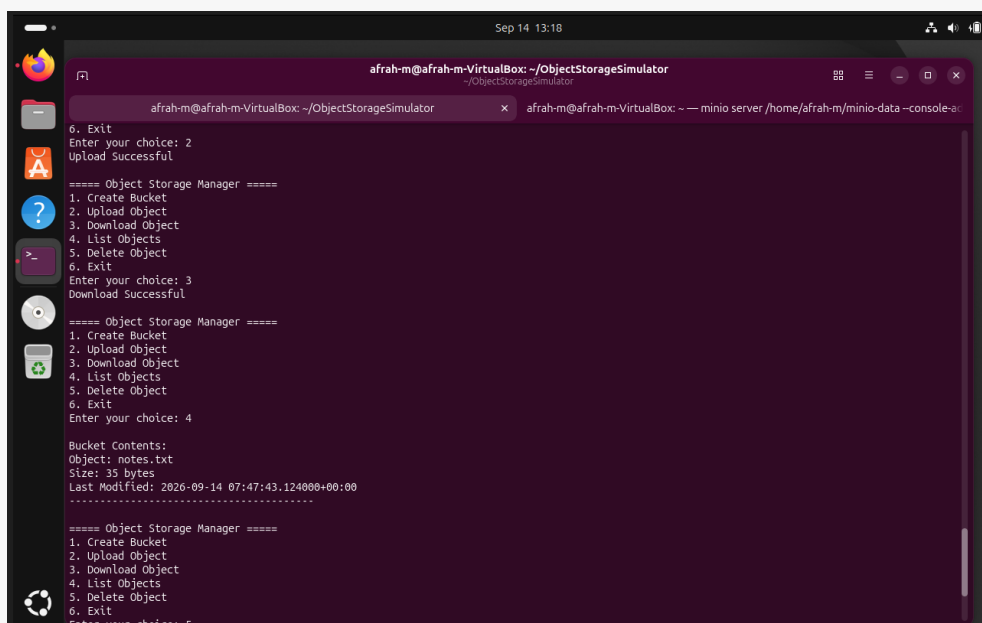


```
afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator
afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator x afrah-m@afrah-m-VirtualBox: ~ — minio server /home/afrah-m/minio-data --console-addr 9000

(venv) afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator$ python3 download.py
Download Successful
(venv) afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator$ cat downloaded.txt
Hello this is sample minio project
(venv) afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator$ nano list_objects.py
(venv) afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator$ python list_objects.py
Object Name: notes.txt
Size: 35 bytes
Last Modified: 2026-09-14 07:42:50.762000+00:00
(venv) afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator$ nano delete_object.py
(venv) afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator$ python delete_object.py
Object Deleted Successfully
(venv) afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator$ python list_objects.py
(venv) afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator$ nano object_storage_manager.py
(venv) afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator$ python object_storage_manager.py

===== Object Storage Manager =====
1. Create Bucket
2. Upload Object
3. Download Object
4. List Objects
5. Delete Object
6. Exit
Enter your choice: 1
Bucket Already Exists

===== Object Storage Manager =====
1. Create Bucket
2. Upload Object
3. Download Object
4. List Objects
5. Delete Object
6. Exit
Enter your choice: 2
Upload Successful
```



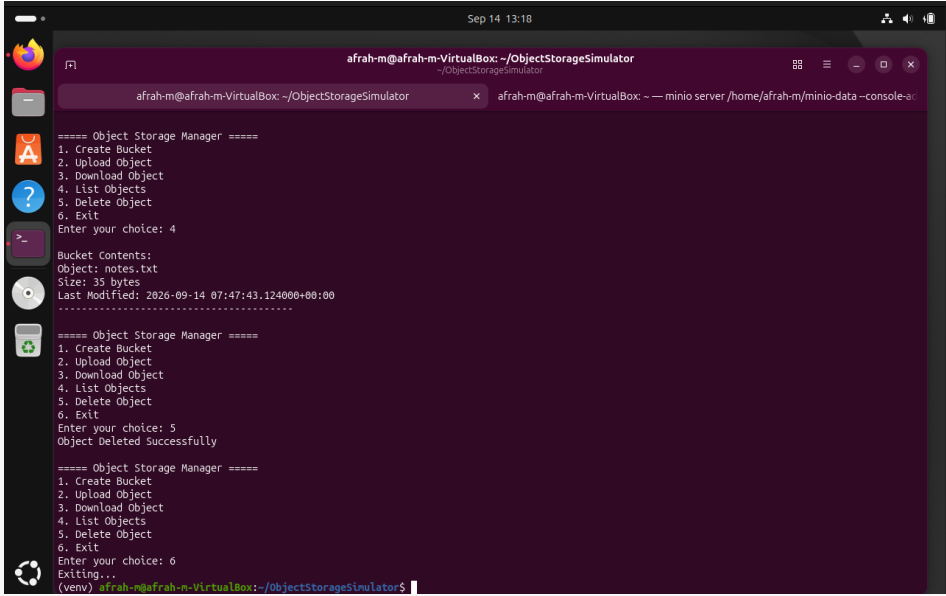
```
6. Exit
Enter your choice: 2
Upload Successful

===== Object Storage Manager =====
1. Create Bucket
2. Upload Object
3. Download Object
4. List Objects
5. Delete Object
6. Exit
Enter your choice: 3
Download Successful

===== Object Storage Manager =====
1. Create Bucket
2. Upload Object
3. Download Object
4. List Objects
5. Delete Object
6. Exit
Enter your choice: 4

Bucket Contents:
Object: notes.txt
Size: 35 bytes
Last Modified: 2026-09-14 07:47:43.124000+00:00
=====

===== Object Storage Manager =====
1. Create Bucket
2. Upload Object
3. Download Object
4. List Objects
5. Delete Object
6. Exit
Enter your choice: 5
```



The screenshot shows a terminal window titled 'afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator'. The terminal displays the following sequence of commands and outputs:

```
afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator
==== Object Storage Manager ====
1. Create Bucket
2. Upload Object
3. Download Object
4. List Objects
5. Delete Object
6. Exit
Enter your choice: 4

Bucket Contents:
Object: notes.txt
Size: 35 bytes
Last Modified: 2026-09-14 07:47:43.124000+00:00
-----

==== Object Storage Manager ====
1. Create Bucket
2. Upload Object
3. Download Object
4. List Objects
5. Delete Object
6. Exit
Enter your choice: 5
Object Deleted Successfully

==== Object Storage Manager ====
1. Create Bucket
2. Upload Object
3. Download Object
4. List Objects
5. Delete Object
6. Exit
Enter your choice: 6
Exiting...
(venv) afrah-m@afrah-m-VirtualBox: ~/ObjectStorageSimulator$
```

## Result

A local S3-compatible object storage environment was successfully created. The Python application performed bucket creation, object upload, download, listing with metadata, and deletion.